

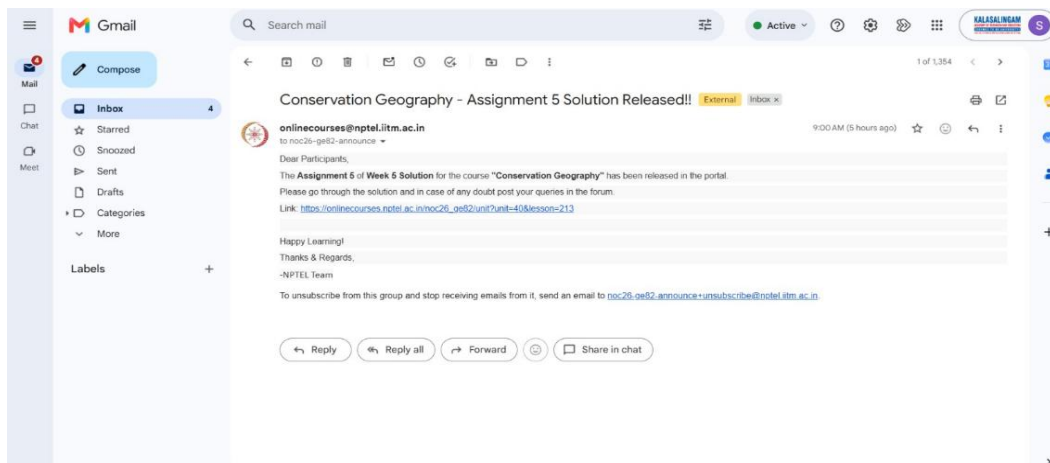
Ex. No 4: Analyze email headers and detect email spoofing using MHA (Mail Header Analyzer)

Description:

Here's a step-by-step guide on how to execute and analyze an email header:

1. Access the Email Header

- Gmail: Open the email, click the three dots (More) in the upper right corner, and select "Show original."
- Outlook: Open the email, click on "File," then "Properties," and find the "Internet headers" box.
- Yahoo: Open the email, click on the three dots (More), and select "View raw message."



2. Copy the Email Header

- Once you access the header, copy all the text that appears. This text contains various metadata fields about the email's journey from the sender to your inbox.

Original Message

Message ID	<Wc6vuGsXQV2NOdmWSxN5vw@geopod-ismtpd-2>
Created at:	Thu, Aug 27, 2026 at 9:00 AM (Delivered after 8 seconds)
From:	onlinecourses@nptel.iitm.ac.in
To:	noc26-ge82-announce@nptel.iitm.ac.in
Subject:	Conservation Geography - Assignment 5 Solution Released!!
SPF:	PASS with IP 209.85.220.69 Learn more
DKIM:	'PASS' with domain nptel-iitm-ac-in.20251104.gappssmtp.com Learn more
DMARC:	'PASS' Learn more

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3. Identify Key Header Fields

- From: The sender's email address.
- To: The recipient's email address.
- Date: The date and time the email was sent.
- Subject: The email's subject line.
- Return-Path: The return address if the email bounces.
- Received: A list of servers the email passed through on its way to the recipient. This is usually the most critical part of header analysis.
- Message-ID: A unique identifier for the email.
- SPF (Sender Policy Framework) / DKIM (DomainKeys Identified Mail): Indicators of email authentication.

4. Analyze the 'Received' Fields

- The Received fields show the path the email took from the sender to the recipient. These fields are listed in reverse order (from the last server to the first).
- Each Received line will show:
 - o The sending server's hostname/IP address.
 - o The receiving server's hostname/IP address.
 - o The date and time the email passed through.

The screenshot shows the MXToolbox 'Analyze Headers' tool interface. At the top, there's a navigation bar with links like Pricing, Tools, Delivery Center, Monitoring, Products, Resources, and Login. Below this is a dark navigation bar with various tool categories, and 'Analyze Headers' is currently selected. The main content area shows the results of an email header analysis for the subject 'Conservation Geography - Assignment 5 Solution Released!!'. A yellow warning box states: 'Copy/Paste Warning: Copy/Pasting a header works for most people, but sometimes it can cause problems with things like DKIM Validation. For the best results, use our Email Deliverability tool'. Under the 'Delivery Information' section, a list of authentication results is shown: DMARC Compliant (green), SPF Alignment (green), SPF Authenticated (green), DKIM Alignment (red), and DKIM Authenticated (green). The 'Relay Information' section shows a 'Received' status and a 'Delay' of 7 seconds.

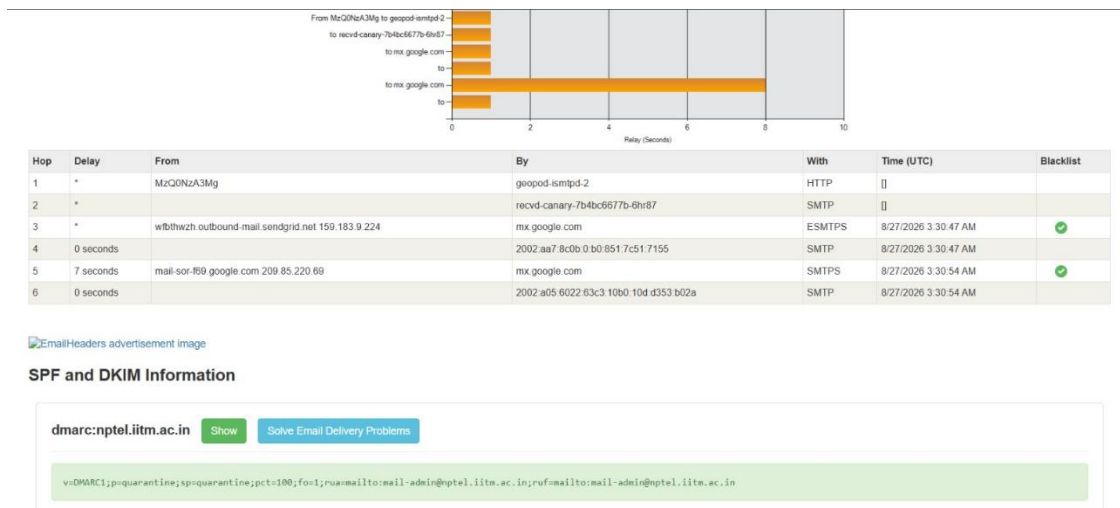
5. Check for IP Addresses and Hostnames

- Use tools like WHOIS or online IP lookup services to identify the geographical location and ownership of the IP addresses found in the Received lines.

- Check if any IP addresses are suspicious or if the hostname does not match the expected sending server.

6. Examine the SPF, DKIM, and DMARC Results

- SPF: Checks if the sender's IP is authorized to send emails for the domain.
 - Pass: Indicates the IP is authorized.
 - Fail: The IP is not authorized and could be a sign of spoofing.
- DKIM: Verifies that the email content has not been altered in transit.
 - Pass: The email has not been tampered with.
 - Fail: The email content may have been altered.
- DMARC: A policy that uses SPF and DKIM to authenticate the email.
 - Pass: Indicates alignment with SPF/DKIM policies.
 - Fail: Indicates possible spoofing or tampering.



7. Analyze the 'Message-ID'

- The Message-ID is unique to each email. It usually includes the domain of the sending server.
- Check if the domain in the Message-ID matches the sender's domain. If not, it could be a sign of spoofing.

8. Look for Anomalies

- Mismatch in Email Domains: Check if the From domain matches the domains in ReturnPath, Message-ID, and other authentication results.
- Strange IP Addresses: Be wary of IP addresses that do not match the known sending server or are from unexpected locations.

- Suspicious Time Stamps: Check for significant time discrepancies in the Received lines, which could indicate delays or rerouting.

9. Use Online Tools

- Use online email header analysis tools like MXToolbox or Google's G Suite Toolbox to automatically parse and analyze the email header.

10. Document and Report Findings

- Document all your findings, especially if you detect phishing or spoofing.
- Report suspicious emails to your IT department or email provider for further investigation.

Headers Found

Header Name	Header Value
Delivered-To	99240041228@klu.ac.in
X-Forwarded-Encrypted	i=4, AHgh+RoBj++VqQ1NPTVeqYMVXoHvD1yQgqh3qIP07B0ARKkzJSXeON8mZn7ZURFTAd1UhaM7AnQw==@klu.ac.in
X-Received	by 2002 a05 6a00 8d86 b0 848 2eac bfb2 with SMTP id d2e1a72cca58-853748a0551mr23974470b3a 13.1787801454456, Wed, 26 Aug 2023 20:30:54 -0700 (PDT)
ARC-Seal	i=3, a=rsa-sha256, t=1787801454, cv=pass, d=google.com, s=arc-20260327, b=FxHvdulkeN1xNPq/4NH8LmrIT3VZNeK11NVL4I2Mws0NoOHLfXJRk0ba50aE5G 1MghHBRlTW8VzqtSIObb8aPah6DSQHngGS9i7zySLs NFjhALegV132UJQJSMIT9 JT1De1mkYgdmMy0DRl77ZxrtYMuCgmMhFDAAV++8uNvryq7T4asPetrG0d9fCEq8 30uVtaxbhwZCR1p0R2G29LGAZCpWVGp/heA06OcdogmuAcr4aro4VgFw/aSove rvz9W9MsZ9/Tnr Y8GzCvmdgkyKq7WqCqCbcobXh+XK09CDJ8xtY0o1yGevK6jZcdv WTRg==
ARC-Message-Signature	i=3, a=rsa-sha256, c=relaxed/relaxed, d=google.com, s=arc-20260327, h=list-unsubscribe list-archive list-help list-post list-id mailing-list precedence to subject message-id mime-version from : date content-transfer-encoding dkim-signature, bh=f2FFQgMbcztD43U7VeD0TdWAEuJ4dwqH4RQy4ns=, fh=PRJUBB71obMdaDlIf4ObdnqB8F9dISNvHlab5CuCW8=, b=Rabdu8+khqhY0Jbc3gl7B6eX7m3TswukTJIA/Z+rwWUpYmIBX8dLZphvZuV0 5Fo240Ghwcpu5QnrUx0hNvSUmID+XIQOpk4BPYvQ1jBSq6lJUSEKdBNtyqswSQ OvOxkS31TvcxFTS3x7otBaNE8DrgpoohtuadYGgp0BDcuEo49eA4ELZ06NjU3q uH9s9Pw9fko8j0avcxOhvy7cRyAdqgV0gXdcZ qNemwE0t8xUqS5Bu0S5E0vXSD ylgUyWhzrEomWAZHf8KsknCU6UINxVpqn9fhdtyWpdDoGhyGL15BFvckSP5uYj1L OK+A==, dara=google.com
ARC-Authentication-Results	i=3, mx.google.com, dkim=pass header i=@ntel-itm.ac.in 20251104 gapssmtp.com header s=20251104 header b=T9M0S3h, arc=pass (i=2 spf=pass spfdomain=mail.ntel.itm.ac.in dkim=pass dkdomain=ntel.itm.ac.in dmarc=pass fromdomain=ntel.itm.ac.in), spf=pass (google.com: domain of noc26-ge82-announce+bnbc25xnbuob2g6x3kamgqehdcredq@ntel.itm.ac.in designates 209 85 220 69 as permitted sender) smtp.mailfrom=noc26-ge82-announce+bnbc25xnbuob2g6x3kamgqehdcredq@ntel.itm.ac.in, dmarc=pass (p=QUARANTINE sp=QUARANTINE dis=NONE) header from=ntel.itm.ac.in, dara=neutral header i=@klu.ac.in
Return-Path	<noc26-ge82-announce+bnbc25xnbuob2g6x3kamgqehdcredq@ntel.itm.ac.in>
Received-SPF	pass (google.com: domain of noc26-ge82-announce+bnbc25xnbuob2g6x3kamgqehdcredq@ntel.itm.ac.in designates 209 85 220 69 as permitted sender) client-ip=209 85 220 69;
Authentication-Results	mx.google.com, dkim=pass header i=@ntel-itm.ac.in 20251104 gapssmtp.com header s=20251104 header b=T9M0S3h, arc=pass (i=2 spf=pass spfdomain=mail.ntel.itm.ac.in dkim=pass dkdomain=ntel.itm.ac.in dmarc=pass fromdomain=ntel.itm.ac.in), spf=pass (google.com: domain of noc26-ge82-announce+bnbc25xnbuob2g6x3kamgqehdcredq@ntel.itm.ac.in designates 209 85 220 69 as permitted sender) smtp.mailfrom=noc26-ge82-announce+bnbc25xnbuob2g6x3kamgqehdcredq@ntel.itm.ac.in, dmarc=pass (p=QUARANTINE sp=QUARANTINE dis=NONE) header from=ntel.itm.ac.in, dara=neutral header i=@klu.ac.in
DKIM-Signature	v=1, a=rsa-sha256, c=relaxed/relaxed, d=ntel-itm.ac.in 20251104 gapssmtp.com, s=20251104, t=1787801454, x=1788406254, dm=klu.ac.in, h=list-unsubscribe list-archive list-help list-post list-id mailing-list precedence: x-original-authentication-results x-original-sender to subject message-id mime-version from : date content-type content-transfer-encoding from to cc subject date message-id reply-to content-type, bh=f2FFQgMbcztD43U7VeD0TdWAEuJ4dwqH4RQy4ns=, b=T9M0S3h3hPUOYfYLYnzSogltS+WGxdlLgBV3WdFXXp+OS0AJNaMqAERTE0i CvFwSqvTn8fK6Gdzunux8k3j7RyTL6wk7my1wUjacrQ4UWZ/CTX0KIR0YjEeth Q0H3ubcuP4FNmerrtGxL+RkH1PH6043is2MIG80zqTK46ASQLcu08KKCOewYzi gW4UBfABIO/WdtxrBMORZIF6jv9+O+Ju6nqSnRdV26DlItg8m504F1jEBvSrdG+Jbn1VFXAZdxoxwmREkvy/8K0dkZQ0/GJ.Snp+vve03yubXyPhPTIShICjI7lBZNK7z h9Fg==
X-Google-DKIM-Signature	v=1, a=rsa-sha256, c=relaxed/relaxed, d=1e100.net, s=20251104, t=1787801454, x=1788406254, h=list-unsubscribe list-archive list-help list-post x-spam-checked-in-group list-id mailing-list precedence x-original-authentication-results x-original-sender to subject message-id mime-version from : date content-type content-transfer-encoding x-beenthere x-gm-message-state from to cc subject date message-id reply-to content-type, bh=f2FFQgMbc

Example of a Header Analysis

Here's a quick example with a sample header: vbnet Copy code

Received: from mail.example.com (mail.example.com [192.0.2.1]) by mail.receiver.com with ESMTP id u29si8604336pjs.40.2023.08.10.07.00.16;

Thu, 10 Aug 2023 07:00:16 -0700 (PDT)

Received: by mail.example.com with SMTP id a1mr1243772ywh.51;

Thu, 10 Aug 2023 07:00:15 -0700 (PDT)

Message-ID: <CA+7eu=4pSeXgQ@mail.example.com>

- Check IP Addresses: Lookup 192.0.2.1 and ensure it is associated with mail.example.com.
- Verify SPF/DKIM: Check if mail.example.com is authorized to send emails for this domain.
- Compare Time Stamps: Ensure the times make sense chronologically without unusual delays.



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[MX Lookup](#)
[Blacklists](#)
[DMARC](#)
[Diagnostics](#)
[Email Health](#)
[DNS Lookup](#)
[Analyze Headers](#)



Email Header Analyzer

Paste Header:

```

!important;border-width:0 !important;margin-top:0 !important;margin-bottom:
0 !important;margin-right:0 !important;margin-left:0 !important;padding-top=
:0 !important;padding-bottom:0 !important;padding-right:0 !important;paddin=
g-left:0 !important;"/>

<p></p>

To unsubscribe from this group and stop receiving emails from it, send an e=
mail to <a href=3D"mailto:noc26-ge82-announce+unsubscribe@nptel.ilitm.ac.in"=
>noc26-ge82-announce+unsubscribe@nptel.ilitm.ac.in</a>.<br />

```

Analyze Header

 Loading

ABOUT EMAIL HEADERS

This tool will make email headers human readable by parsing them according to RFC 822. Email headers are present on every ema results and more. If you need help getting copies of your email headers, [just read this tutorial](#).

Rubrics

Criteria & Marks Assigned	Mark Allotted	Mark Awarded
1. GitHub Activity & Submission Regularity	3	
2. Application of Forensic Tools & Practical Execution	3	
3. Documentation & Reporting	2	
4. Engagement, Problem-Solving & Team Collaboration	2	
Total	10	

Result

The email header was successfully extracted and analyzed using MXTToolbox. Authentication protocols (SPF, DKIM, DMARC) and the originating IP address were manually verified, demonstrating how to trace email origins and detect potential spoofing.